Supplemental Digital Content 3

Calibration summary — full bisection and ABC-SMC results including all rate constants and credible intervals.

This table provides the complete calibration parameters referenced in Sections 2.8 and 3.1 of the main manuscript. Calibration anchors the per-outcome hazard constants to the pooled Colombian Aerospace Force (FAC) DIMAE 2010–2026 registry (n = 7,271 chamber exposures, n = 173 clinical middle-ear barotrauma events).

Metric	Value
Target per-exposure pbarotitis (FAC pooled 2010–2026)	2.38%
Wilson 95% confidence interval on target	[2.06%, 2.75%]
Achieved per-exposure pbarotitis (bisection calibrator)	2.47%
Calibration residual (achieved − target)	+0.09 percentage points
Career projection (3 exposures, simulated)	7.23%
Pooled FAC 2010–2026 career anchor	6.97%
Career-3 residual (simulated − anchor)	+0.26 percentage points
rbarotitis (bisection point estimate; per-second hazard rate constant)	5.85×10^{-8}
rbaromyringitis (bisection point estimate)	1.75×10^{-9}
rrupture (bisection point estimate)	5.85×10^{-11}
Bisection iterations to convergence	6
ABC-SMC posterior mean rbarotitis (fit against 2.00% target)	5.07×10^{-8}
ABC-SMC 95% credible interval on rbarotitis	[3.59×10^{-8}, 8.00×10^{-8}]
Bisection re-anchored estimate (5.85×10^{-8}) inside ABC-SMC 95% CrI	Yes

ABC-SMC = Approximate Bayesian Computation Sequential Monte

Carlo (reference 19 in main manuscript). Run configuration: 100 particles, 4 generations, 150 synthetic-cohort subjects per particle, tolerance schedule 13.9 → 5.0 against three summary statistics (cohort prevalence; URI day 4–7 / none gradient; severe / normal severity gradient).